Mario Torres Danta

mtdanta[at]outlook.com
LinkedIn: mtdanta
Web: mtdanta.github.io
Seville, Spain

Education

B.Sc. in Mathematics *2023 – 2027 (Expected)*

University of Seville (US), Seville, Spain

Current GPA: 8.74 / 10.0 (Top 5% of cohort; based on 22 graded courses)

Academic Honors and Distinctions

Collaboration Grant (*Beca de Colaboración*), Spanish Government *2026 – 2027*

Competitive national research-initiation grant for top final-year students, held in a university department.

Awarded **"Matrícula de Honor" (Highest Distinction)** in several courses, such as:

- **General Topology (10.0)**
- **Measure Theory & Lebesgue Integration (10.0)**
- **Mathematical Optimization (9.2)**
- **Numerical Analysis I (9.5)**
- **Ordinary Differential Equations (9.4)**
- **Probability Theory (9.4)**

Additional "Sobresaliente" (A-Grade) in courses such as Local Geometry of Surfaces (10.0), Algebraic Structures, Complex Analysis, Linear Algebra II, Discrete Mathematics.

Research Experience

Visiting Student & Undergraduate Researcher *Aug 2026 – Sep 2026*

University of Copenhagen, Denmark

Project: Research project on mapping class groups of surfaces, Teichmüller theory and Riemann moduli spaces, including braid groups and homological stability. Resulted in a manuscript in preparation (see below).

Supervisor: Prof. Nathalie Wahl

. . .

FUSRP Undergraduate Researcher *Jun 2026 – Jul 2026*

The Fields Institute for Research in Mathematical Sciences, Toronto, Canada

Project: Laurent symmetric functions and their relation with the representation theory of general linear groups. Resulted in a preprint (see below).

Supervisors: Prof. Alistair Savage and Dr. Yaolong Shen

. . .

Undergraduate Researcher (Departmental Collaborator) *Sep 2025 – Jun 2026*

Department of Geometry and Topology, University of Seville

Project: Advanced topics in Topology and Group Theory. A guided study complementing the standard undergraduate curriculum.

Supervisor: Prof. Ramón J. Flores

. . .

Undergraduate Researcher (Departmental Collaborator) *Sep 2024 – Jun 2025*

Department of Algebra, University of Seville

Project: Classification of homographies and homologies in real ($\mathbb{P}^n(\mathbb{R})$) and complex ($\mathbb{P}^n(\mathbb{C})$) projective spaces.

Supervisor: Prof. Francisco Castro Jiménez

Publications and Manuscripts

H. Fernandez, E. Kleinwort, T. Li and **M. Torres-Danta**, *Laurent symmetric functions*, preprint (2026). arXiv:2609.15690.

M. Torres-Danta and N. Wahl, *The symmetric mapping class group of finite abelian branched covers*, manuscript in preparation (2026).

Research Interests

Geometric Topology (Mapping Class Groups, Braid Groups) · Algebraic Topology (Homology, Homotopy Theory, Homological Stability) · Group Theory · Representation Theory · Category Theory

Academic Service

Reviewer, zbMATH Open *Sep 2026 – Present*

Writing reviews of recent research articles for the zbMATH Open database.

Academic Engagement

Workshop on Geometric Topology *(informal workshop)* *Feb 2026*

Role: Attendee. Sessions on geometric topology and related structures. [View Website]

. . .

ByMat Conference (Bringing Young Mathematicians Together) *Nov 2025*

University of Seville / RSME (Royal Spanish Mathematical Society)

Role: Attendee. Selected to participate in this conference for young researchers.

Highlight: Particular interest in plenary lectures by **Prof. Marithania Silvero** (Knot Theory) and **Prof. Alberto Rodríguez** (Differential Geometry/Topology), aligning with my research focus.

. . .

Independent Advanced Study *2025 – Present*

Self-study of graduate-level texts to bridge the gap between undergraduate topology and current research:

- **B. Farb & D. Margalit, *A Primer on Mapping Class Groups*:** Surface homeomorphisms, Dehn twists and braid groups (preparation for Copenhagen).
- **J. Munkres, *Topology* & *Elements of Algebraic Topology*:** Fundamental groups and covering spaces; currently studying singular homology.
- **S. Kalajdzievski, *An Illustrated Introduction to Topology and Homology*:** Visualization techniques for higher-dimensional structures.

Skills

- **Languages:** Spanish (Native), English (C1), French (B2)
- **Programming: Python** (Proficient; *Santander Open Academy* certificate), **MATLAB**
- **Technical: LaTeX** (Primary document preparation), **GitHub** (Repository management)

References

Prof. Nathalie Wahl
Professor, Dept. of Mathematical Sciences, University of Copenhagen
E-mail: wahl@math.ku.dk
Web: web.math.ku.dk/~wahl

Prof. Alistair Savage
Professor, Dept. of Mathematics and Statistics, University of Ottawa
E-mail: alistair.savage@uottawa.ca
Web: alistairsavage.ca

Prof. Ramón J. Flores
Professor, Dept. of Geometry and Topology, University of Seville
E-mail: ramonjflores@us.es
Web: personal.us.es/ramonjflores

Full academic transcript available upon request.